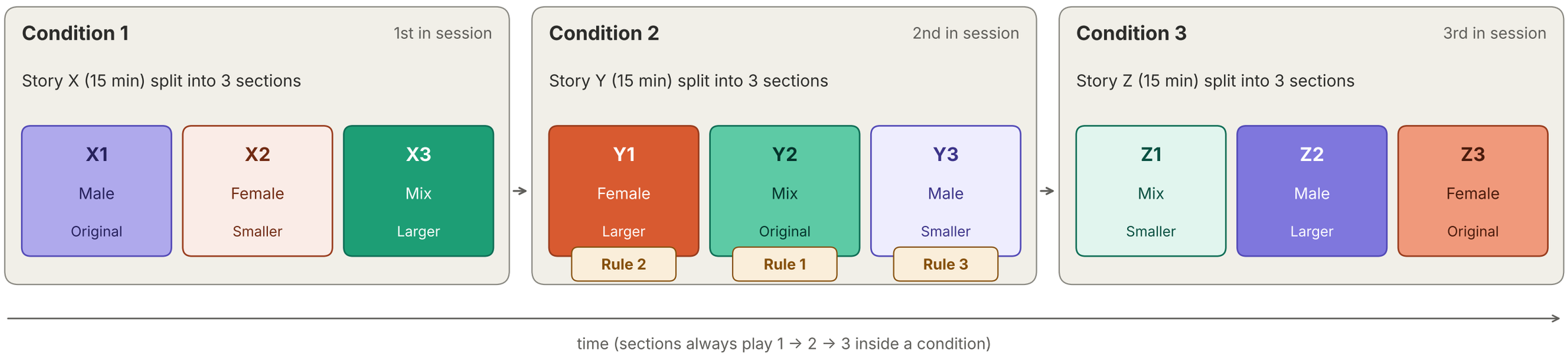
Experimental design: conditions, stories, voices and rules

Within-subjects. Each participant hears 3 conditions × 3 five-minute sections = 9 sections (45 min).

A One session (example: participant 1)



What is counterbalanced

- **Condition order** — All 6 orders of C1/C2/C3, rotated across participants.
- **Story ↔ condition** — Each condition plays one whole story; stories rotate (Latin square), so every story meets every condition and every session position.
- **Section order** — Fixed: 1 → 2 → 3 (the story must run in order).
- **Voice per section** — Within every condition: all 3 genders and all 3 variants, one per section. All 9 voices used once per participant.
- **Condition 2 rules** — Each section of C2 carries one rule; the rule order (6 possible) rotates so every rule meets every section, gender, variant and story equally.

Legend Male Female Mix — shade: mid = original · dark = larger/darker · light = smaller/brighter

B Assignment for all 18 participants (repeat the cycle every 18)

P	Session order	Condition 1			Condition 2			Condition 3		
		sec 1	sec 2	sec 3	sec 1	sec 2	sec 3	sec 1	sec 2	sec 3
1	C1 → C2 → C3	X1	X2	X3	Y1 · R2	Y2 · R1	Y3 · R3	Z1	Z2	Z3
2	C1 → C3 → C2	Y1	Y2	Y3	Z1 · R2	Z2 · R1	Z3 · R3	X1	X2	X3
3	C2 → C1 → C3	Z1	Z2	Z3	X1 · R3	X2 · R2	X3 · R1	Y1	Y2	Y3
4	C2 → C3 → C1	Y1	Y2	Y3	Z1 · R3	Z2 · R1	Z3 · R2	X1	X2	X3
5	C3 → C1 → C2	Z1	Z2	Z3	X1 · R2	X2 · R3	X3 · R1	Y1	Y2	Y3
6	C3 → C2 → C1	X1	X2	X3	Y1 · R1	Y2 · R2	Y3 · R3	Z1	Z2	Z3
7	C1 → C2 → C3	Z1	Z2	Z3	X1 · R3	X2 · R2	X3 · R1	Y1	Y2	Y3
8	C1 → C3 → C2	X1	X2	X3	Y1 · R1	Y2 · R2	Y3 · R3	Z1	Z2	Z3
9	C2 → C1 → C3	Y1	Y2	Y3	Z1 · R2	Z2 · R3	Z3 · R1	X1	X2	X3
10	C2 → C3 → C1	X1	X2	X3	Y1 · R1	Y2 · R3	Y3 · R2	Z1	Z2	Z3
11	C3 → C1 → C2	Y1	Y2	Y3	Z1 · R1	Z2 · R3	Z3 · R2	X1	X2	X3
12	C3 → C2 → C1	Z1	Z2	Z3	X1 · R3	X2 · R1	X3 · R2	Y1	Y2	Y3
13	C1 → C2 → C3	Y1	Y2	Y3	Z1 · R2	Z2 · R3	Z3 · R1	X1	X2	X3
14	C1 → C3 → C2	Z1	Z2	Z3	X1 · R3	X2 · R2	X3 · R1	Y1	Y2	Y3
15	C2 → C1 → C3	X1	X2	X3	Y1 · R2	Y2 · R1	Y3 · R3	Z1	Z2	Z3
16	C2 → C3 → C1	Z1	Z2	Z3	X1 · R1	X2 · R2	X3 · R3	Y1	Y2	Y3
17	C3 → C1 → C2	X1	X2	X3	Y1 · R3	Y2 · R1	Y3 · R2	Z1	Z2	Z3
18	C3 → C2 → C1	Y1	Y2	Y3	Z1 · R1	Z2 · R3	Z3 · R2	X1	X2	X3

Cell text: story + section (e.g. Y2 = Story Y, section 2); in Condition 2, R = rule. Colour = voice gender, shade = voice variant.

Balance over 18 participants: each story × condition 6×; each story and condition in each session position 6×; every story section heard in all 9 voices (2× each);

Condition 2: each rule × section, × gender, × variant and × story 6×; each of the 6 rule orders used 3× (once per C2 session position).